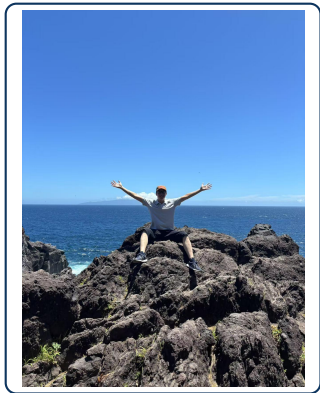


Stationary MMD Points

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Research Interests

- Kernel and nonparametric methods
- Causal inference
- Statistical learning theory

Motivation

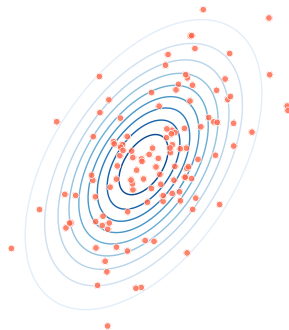
Problem: Approximation of a target distribution μ with a finite set of points $\{\mathbf{x}_i\}_{i=1}^n$.

- Denote the empirical distribution of $\{\mathbf{x}_i\}_{i=1}^n$ as

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{\mathbf{x}_i}.$$

- We want $\hat{\mu}_n$ to be close to μ in some sense.

$$\lim_{n \rightarrow \infty} D(\hat{\mu}_n, \mu) = 0.$$

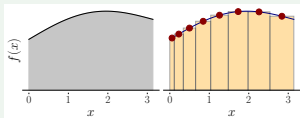


Motivation

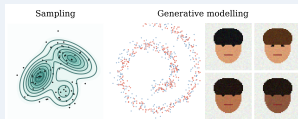
Problem: Approximation of a target distribution μ with a finite set of points $\{\mathbf{x}_i\}_{i=1}^n$.



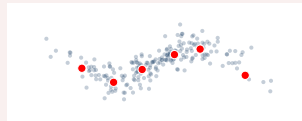
Numerical Integration



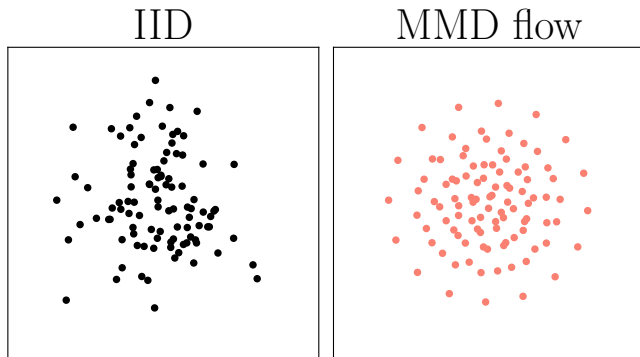
Sampling



Coreset Selection



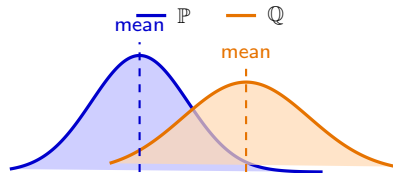
An empirical observation



- Samples from MMD flow are more **uniform** than IID samples [XKS22].

Background: What is MMD?

- Given two distributions $\mathbb{P}$ and $\mathbb{Q}$, how to compare them?
- $|\mathbb{E}_{\mathbb{P}}[X] - \mathbb{E}_{\mathbb{Q}}[X]|$.
- $|\mathbb{E}_{\mathbb{P}}[X^2] - \mathbb{E}_{\mathbb{Q}}[X^2]|$.
- The first two moments are not enough!



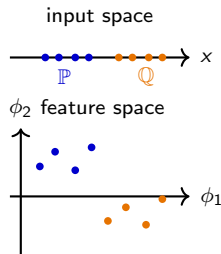
Goal: Compare $\mathbb{P}$ and $\mathbb{Q}$ with a general f , i.e., $|\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|$?

Background: Maximum Mean Discrepancy (MMD) [GBR⁺12]

- $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS) with kernel k .

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- The RKHS contains functions that are linear combinations of the feature maps $\phi : \mathcal{X} \rightarrow \mathcal{H}$, $f = \sum_{i=1}^n \alpha_i \phi(x_i)$.



Theorem

MMD admits a closed-form expression:

$$\text{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{X, X' \sim \mathbb{P}}[k(X, X')] + \mathbb{E}_{Y, Y' \sim \mathbb{Q}}[k(Y, Y')] - 2\mathbb{E}_{X \sim \mathbb{P}, Y \sim \mathbb{Q}}[k(X, Y)].$$

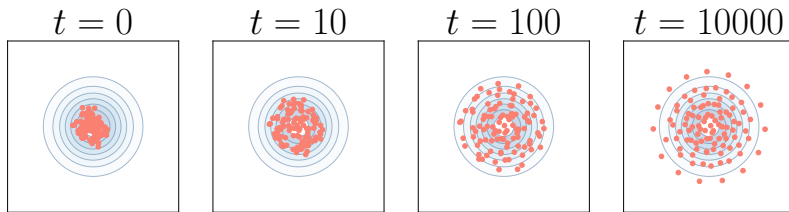
- MMD is a probability divergence ($\text{MMD}(\mathbb{P}, \mathbb{Q}) = 0 \iff \mathbb{P} = \mathbb{Q}$) if k is characteristic (Gaussian) [SFL11].

Background: MMD gradient flow [AKSG19]

- Start with an initial set of n particles $\{\mathbf{x}_i^{(0)}\}_{i=1}^n$ at time 0.
- Evolve the particles according to the MMD gradient flow:

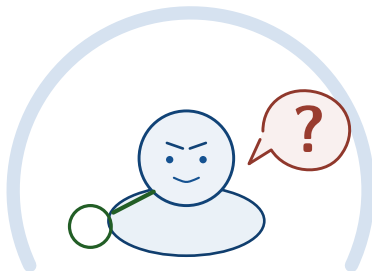
$$\mathbf{x}_i^{(t+1)} = \mathbf{x}_i^{(t)} - \gamma \cdot \nabla_{\mathbf{x}_i^{(t)}} \text{MMD}^2 \left(\mu, \frac{1}{n} \sum_{j=1}^n \delta_{\mathbf{x}_j^{(t)}} \right), \quad i = 1, \dots, n.$$

- $\gamma > 0$ is the step size.



- A Wasserstein gradient flow.

Observation: MMD flow samples are more **uniform** than IID samples [XKS22].



Existing work

- Existing finite-sample rates in MMD gradient flows [AKSG19, CMG⁺25]
- Let $\mu_n^{(t)} = \frac{1}{n} \sum_{j=1}^n \delta_{\mathbf{x}_j^{(t)}}$ be the empirical distribution of n particles at iterate t .

$$\text{MMD} \left(\mu, \mu_n^{(t)} \right) \leq C(t) n^{-\frac{1}{2}}, \quad W_2 \left(\mu, \mu_n^{(t)} \right) \leq C(t) n^{-\frac{1}{2}}.$$

Two limitations

- Same rate as IID sampling: $n^{-\frac{1}{2}}$.
- The constant $C(t)$ grows exponentially in t .

- Minimum MMD points

$$\{\mathbf{x}_i^*\}_{i=1}^n = \arg \min_{\{\mathbf{x}_i\}_{i=1}^n} \text{MMD} \left(\mu, \frac{1}{n} \sum_{j=1}^n \delta_{\mathbf{x}_j} \right).$$

- Minimum MMD points achieve better than IID rate [XKS22, DP10].
- Let $\mu_n^* = \frac{1}{n} \sum_{j=1}^n \delta_{\mathbf{x}_j^*}$ be the empirical distribution of n minimum MMD points.

$$\text{MMD}(\mu, \mu_n^*) \leq Cn^{-1}.$$

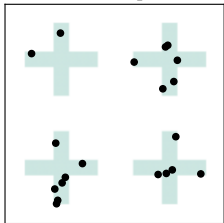
Two limitations

- Minimum MMD points are hard to compute, due to the non-convexity of MMD.
- Gradient descent only finds a **stationary** point instead.

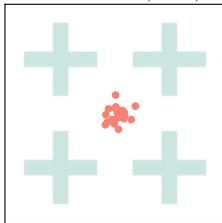
Motivation

× ~~Minimum MMD points~~ but ✓ **Stationary MMD points**

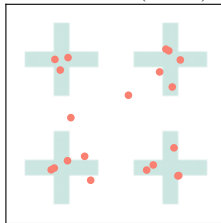
IID Samples



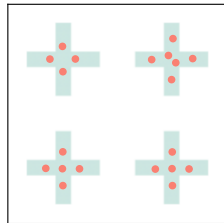
MMD Flow ($T=0$)



MMD Flow ($T=20$)



Final



Stationary MMD points

Definition (Stationary MMD points)

A sequence of point sets $\{x_i\}_{i=1}^n$ with $\mu_n := \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$, that satisfies:

1. **Convergence:** $\text{MMD}(\mu, \mu_n) \rightarrow 0$ as $n \rightarrow \infty$.
2. **Stationarity:** $\nabla_{x_1, \dots, x_n} \text{MMD}^2(\mu, \mu_n) = 0$.

- Minimum MMD points are a special case of stationary MMD points.

Question: Why does stationary MMD points exhibit superior numerical integration performance than IID points?

Stationary MMD points

- Simulate MMD flow for long enough, $t \rightarrow \infty$.
- For any $i \in \{1, \dots, n\}$:

$$\nabla_{x_i} \text{MMD}^2 \left(\mu, \frac{1}{n} \sum_{j=1}^n \delta_{x_j} \right) = 0 \implies \frac{1}{n} \sum_{j=1}^n \nabla_1 k(x_i, x_j) - \int \nabla_1 k(x_i, y) d\mu(y) = 0.$$

- Exact integration on $x \mapsto \nabla_1 k(x_i, x)$.
- Exact integration on $\{1\}$.

Exact integration subspace:

$$\mathcal{F}_n := \text{span} \left(\{1\} \cup \{ \nabla_1 k(x_i, \cdot) : i \in \{1, \dots, n\} \} \right) \subset \mathcal{H}.$$

Stationary MMD points

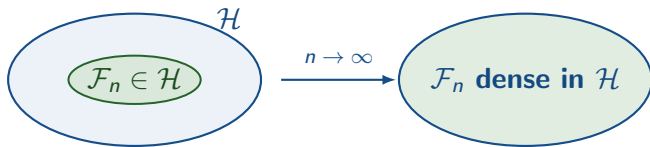
- Suppose the kernel k is continuously differentiable.
- Suppose the kernel k is c_0 -universal.

Proposition (Approximation of $\mathcal{F}_n$ towards $\mathcal{H}$)

Given $f \in \mathcal{H}$. Consider the semi-norm

$$|f|_n := \inf \{ \|r_n\|_{\mathcal{H}} : f = f_n + r_n, f_n \in \mathcal{F}_n, r_n \in \mathcal{H} \}$$

Then, $\lim_{n \rightarrow \infty} |f|_n = 0$.



Stationary MMD points

Theorem (Super-convergence)

For a stationary MMD point set $\{x_i\}_{i=1}^n$ and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$,

$$\left| \frac{1}{n} \sum_{i=1}^n f(x_i) - \int f \, d\mu \right| \leq |f|_n \text{MMD}(\mu, \mu_n).$$

- Since $\underbrace{|f|_n}_{\substack{\rightarrow 0 \\ \text{stationarity}}} \rightarrow 0$, the estimator has error $o\left(\underbrace{\text{MMD}(\mu, \mu_n)}_{\substack{\rightarrow 0 \\ \text{MMD flow}}}\right)$.

Proof sketch

- Decompose $f = f_n + r_n$ with $f_n \in \mathcal{F}_n$.

$$\left| \frac{1}{n} \sum_{i=1}^n f(x_i) - \int f \, d\mu \right| = \left| \frac{1}{n} \sum_{i=1}^n r_n(x_i) - \int r_n \, d\mu \right| \leq \|r_n\|_{\mathcal{H}} \cdot \text{MMD}(\mu, \mu_n),$$

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- [DP10] Josef Dick and Friedrich Pillichshammer. *Digital nets and sequences: discrepancy theory and quasi-Monte Carlo integration*. Cambridge University Press, 2010.
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- [SFL11] Bharath K. Sriperumbudur, Kenji Fukumizu, and Gert R. G. Lanckriet. Universality, characteristic kernels and RKHS embedding of measures. *Journal of Machine Learning Research*, 12(7):2389–2410, 2011.
- [XKS22] Lantian Xu, Anna Korba, and Dejan Slepcev. Accurate quantization of measures via interacting particle-based optimization. In *International Conference on Machine Learning*, pages 24576–24595. PMLR, 2022.